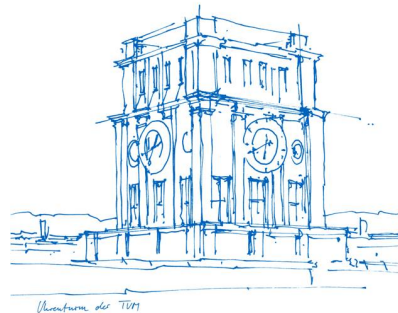


PoreAccess-CO₂: From Random Packing to an Interpretable Transport Closure

Structure-resolved diffusion and adsorption in DEM-generated zeolite beds

Technical University of Munich
Chair of Process Systems Engineering

Ali Khalifa & Heiko Briesen



2026-09-12

Good morning. This project is called PoreAccess-CO₂. It examines how the random arrangement of particles changes CO₂ transport through a packed zeolite bed. The main aim is to retain information about the pore structure while developing a model that is much faster than a complete pore-network adsorption simulation. I will first introduce the physical problem, then explain the pore-network and one-dimensional models. Finally, I will show how we connect them through an accessibility-based transport closure and what the first twenty packing realizations tell us.



Presentation outline

Motivation and research question

Model scope and physical mechanisms

Numerical methods and diffusivity prediction

Results and robustness

Conclusions and next steps



2026-09-12

Presentation outline

The presentation moves from the research question to the physical model, numerical methods, results and conclusions.

Presentation outline

Motivation and research question

Model scope and physical mechanisms

Numerical methods and diffusivity prediction

Results and robustness

Conclusions and next steps

Section 1 of 5

Motivation and research question

Why can similar beds adsorb differently?

Why can similar beds adsorb differently?.

CO₂ adsorption matters and packing structure is a missing scale



Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

What is already established?

- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

2026-09-12

└ Motivation and research question

└ CO₂ adsorption matters and packing structure is a missing scale

CO₂ adsorption matters and packing structure is a missing scale

Why CO₂ capturing matters?

- Food systems contribute about **one-third** of global anthropogenic greenhouse-gas emissions.
- Food and bioprocessing include recoverable biogenic CO₂ streams, for example from fermentation.
- Regenerable sorbents offer a compact capture and purification route.

What is already established?

- CO₂ equilibrium and uptake kinetics on zeolite 13X are widely studied.
- Fixed-bed models normally use bulk porosity and a fitted effective transport coefficient.
- Random inlet-connected paths remain hidden inside that coefficient.

Novelty: connect random particle geometry to a fast bed-scale uptake model through a physically calculated pore-accessibility descriptor.

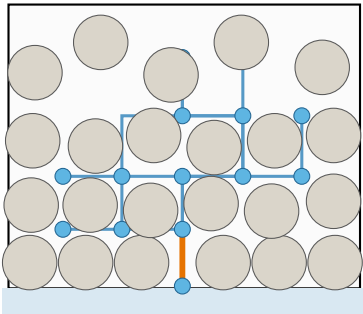
CO₂ capture and separation matter across many sectors. Food systems contribute roughly one third of anthropogenic greenhouse-gas emissions, and food and bioprocessing also generate concentrated biogenic CO₂ streams, for example during fermentation. Adsorption offers one possible route for capturing or purifying such streams.

The equilibrium behaviour and uptake kinetics of CO₂ on zeolite 13X have already received extensive attention. At bed scale, however, models usually represent a random packing only through its average porosity and an effective transport coefficient. That treatment hides whether the void space forms several open paths from the inlet or only a few restricted paths. Our novelty lies in connecting the random particle geometry to a fast bed-scale uptake model through a physically calculated accessibility descriptor.

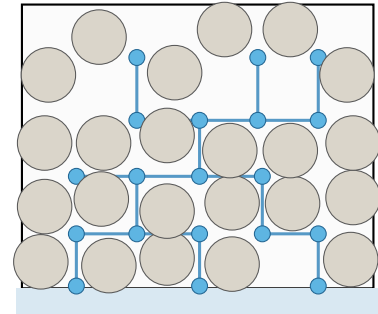
Same porosity can hide different inlet-connected pathways

Can a pore-accessibility measure predict bed-scale transient transport better than porosity alone?

Packing A: inlet bottleneck



Packing B: multiple inlet paths



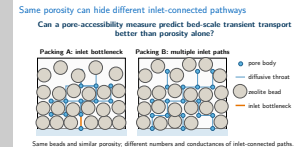
- pore body
- diffusive throat
- zeolite bead
- inlet bottleneck

Same beads and similar porosity; different numbers and conductances of inlet-connected paths.

2026-09-12

└ Motivation and research question

└ Same porosity can hide different inlet-connected pathways



This schematic introduces the central problem. The two beds contain the same monodisperse beads and can have almost the same total porosity. Their inlet-connected paths can nevertheless differ strongly. Packing A illustrates a network reached through one restrictive inlet bottleneck. Packing B illustrates several paths that connect the reservoir to the bed interior.

The blue lines show only simplified inlet-connected paths, not every throat in a real extracted network. The sketches are therefore hypotheses rather than simulation results. They motivate our research question: can a quantitative pore-accessibility measure predict transient bed-scale transport better than bulk porosity alone?

Why zeolite 13X and what does the study isolate?

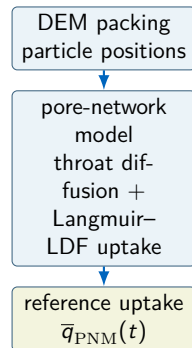
Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Motivation and research question

Why zeolite 13X and what does the study isolate?

We use zeolite 13X because it is a well-established benchmark sorbent for CO₂. It has strong equilibrium uptake at low partial pressure and is available as millimetre-scale beads that naturally form random beds.

The study deliberately isolates one mechanism. The gas is dry, isothermal and stagnant. CO₂ enters from a reservoir below the bed. We do not impose a gas flow and we do not solve CFD. We do, however, model the pore space explicitly. Starting from the DEM particle positions, the pore-network model resolves diffusion through the throats and couples it to Langmuir equilibrium and finite-rate adsorption in the beads. Its principal reference output is the mean network uptake curve, $\bar{q}_{\text{PNM}}(t)$.

Why zeolite 13X?

- established benchmark sorbent for CO₂
- strong equilibrium uptake at low partial pressure
- commercial millimetre-scale beads form random beds

Controlled model problem

- dry, isothermal, stagnant gas
- CO₂ enters from a reservoir below the bed
- pore-network model resolves throat diffusion and bead adsorption
- no imposed flow; no CFD

How strongly does random void connectivity change access to the adsorbent?



Section 2 of 5

Model scope and physical mechanisms

From particle arrangement to diffusion and adsorption

From particle arrangement to diffusion and adsorption.

Twenty beds isolate variability caused by random arrangement



Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m ⁻³
simulation time	5000 s

Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

The ensemble tests arrangement effects under otherwise identical conditions.

2026-09-12

- └ Model scope and physical mechanisms
- └ Twenty beds isolate variability caused by random arrangement

Twenty beds isolate variability caused by random arrangement	
Quantity	Fixed value
beads per bed	1850
bead diameter	1.6 mm
tube diameter	16 mm
D_{tube}/d_p	10
temperature	298.15 K
inlet c_{CO_2}	6.05 mol m ⁻³
simulation time	5000 s

Only the DEM insertion seed changed

- Delaunay–Voronoi networks were built directly from particle coordinates
- all networks were connected and spanning
- quasi-uniform Sobol sampling allocated pore volumes without a voxel grid
- transient and 1D models passed mass-balance checks

The ensemble tests arrangement effects under otherwise identical conditions.

We generated twenty independent beds. Every listed physical and numerical condition remains fixed, including the number and diameter of the beads, tube diameter, temperature, inlet concentration and simulation duration. Only the DEM insertion seed changes. Differences between the cases can therefore be attributed to the random arrangement rather than to changed operating conditions.

We construct a Delaunay–Voronoi pore network directly from the particle coordinates. Sobol sampling means a deterministic, low-discrepancy set of quasi-uniform points. We use these points only to allocate the analytically calculated void volume among pore bodies. They do not create the network connectivity, and no voxel grid is required. All twenty networks span the bed and pass the mass-balance checks.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels

A low-discrepancy Sobol sample of 2^{22} points allocates the analytically calculated void volume to pore bodies; it does not define connectivity.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

2026-09-12

- └ Model scope and physical mechanisms
- └ Pore-network geometry and throat diffusion

Four neighbouring bead centres define a Delaunay tetrahedron, and its circumcentre becomes a pore body. Two tetrahedra sharing a triangular face create two neighbouring pores connected by a throat. The indices i and j therefore refer to adjacent pore bodies. Diffusion through that throat follows $G_{ij} = D_m A_{ij} / L_{ij}$. Here D_m has units of square metres per second, A_{ij} has units of square metres and L_{ij} has units of metres. Thus, G_{ij} has units of cubic metres per second. Multiplying it by the concentration difference gives the molar flow rate through the throat. The actual opening is irregular. We approximate it by a circular area $A_{ij} = \pi r_{ij}^2$. This equivalent circular area is an explicit geometric assumption of the model.

Pore-network geometry and throat diffusion

Network construction

- four bead centres define a Delaunay tetrahedron; its circumcentre is a pore body
- tetrahedra sharing a triangular face are joined by a throat
- topology comes from particle coordinates, not voxels

A low-discrepancy Sobol sample of 2^{22} points allocates the analytically calculated void volume to pore bodies; it does not define connectivity.

Diffusion through throat $i-j$

$$G_{ij} = D_m \frac{A_{ij}}{L_{ij}}, \quad \dot{n}_{ij} = G_{ij}(c_j - c_i).$$

$L_{ij} = \|\mathbf{x}_j - \mathbf{x}_i\|$ is the pore-centre distance. The irregular opening between three beads is approximated by an equivalent circular area

$$A_{ij} = \pi r_{ij}^2,$$

where r_{ij} is the clearance of the shared Delaunay face.

The circular throat area is a geometric modelling assumption; G_{ij} has units of $\text{m}^3 \text{s}^{-1}$.

Langmuir equilibrium and bead uptake	
Local equilibrium loading	Meaning of the rate equation
$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}$	q_k is the current loading of bead k k_{LDF} controls how quickly q_k approaches q_k^* - gas removed from adjacent pores is added to the bead
Finite-rate uptake	Pore-network outputs
$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k)$	- pore concentrations $c_i(t)$ - particle loadings $q_k(t)$ - mean uptake $\bar{q}_{\text{PNM}}(t)$
q_s is the maximum CO_2 loading [mol/kg adsorbent], b is the Langmuir affinity, p_k is the local CO_2 partial pressure, and q_k^* is the corresponding equilibrium loading. The product $b p_k$ is dimensionless. The coupled diffusion-adsorption calculation conserves the total amount of CO_2.	

Langmuir equilibrium and bead uptake

Local equilibrium loading

$$p_k = c_k RT, \quad q_k^* = \frac{q_s b p_k}{1 + b p_k}.$$

Finite-rate uptake

$$\frac{dq_k}{dt} = k_{\text{LDF}} (q_k^* - q_k).$$

q_s is the maximum CO_2 loading [mol/kg adsorbent]. b is the Langmuir affinity, p_k is the local CO_2 partial pressure, and q_k^* is the corresponding equilibrium loading. The product $b p_k$ is dimensionless.

The coupled diffusion-adsorption calculation conserves the total amount of CO_2 .

Meaning of the rate equation

- q_k is the current loading of bead k
- k_{LDF} controls how quickly q_k approaches q_k^*
- gas removed from adjacent pores is added to the bead

Pore-network outputs

- pore concentrations $c_i(t)$
- particle loadings $q_k(t)$
- mean uptake $\bar{q}_{\text{PNM}}(t)$

Diffusion supplies a local gas concentration c_k around bead k . We convert it to partial pressure through $p_k = c_k RT$. The Langmuir equation then gives the equilibrium loading q_k^* . The parameter q_s is the saturation loading and b is the Langmuir affinity. Because b has units of inverse pressure, the product $b p_k$ is dimensionless.

The actual particle loading q_k does not jump immediately to equilibrium. The linear-driving-force equation describes how it approaches q_k^* , with k_{LDF} setting the uptake rate. Gas removed from the pores is added to the particles, so the coupled calculation conserves CO_2 . The detailed outputs include every pore concentration $c_i(t)$ and every bead loading $q_k(t)$. Averaging the bead inventory gives the reference uptake curve $\bar{q}_{\text{PNM}}(t)$.

Section 3 of 5

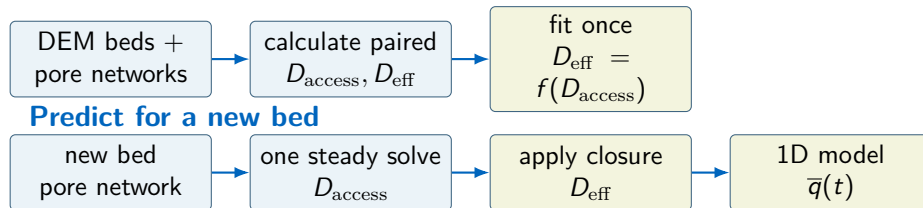
Numerical methods and diffusivity prediction

How we calculate accessibility and calibrate the reduced model

How we calculate accessibility and calibrate the reduced model.

Development and later use are two different workflows

Develop closure



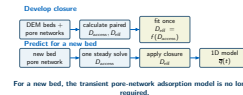
For a new bed, the transient pore-network adsorption model is no longer required.

2026-09-12

└ Numerical methods and diffusivity prediction

└ Development and later use are two different workflows

Development and later use are two different workflows



This slide separates development from later prediction. During development, we run the full transient pore-network adsorption model for each training bed. From its uptake curve, we fit one effective diffusivity, D_{eff} . Independently, one steady network solve gives the accessibility diffusivity, D_{access} . The paired values across the beds allow us to fit the closure $D_{\text{eff}} = f(D_{\text{access}})$. For a new packing, the lower workflow applies. We still extract its pore network, but we perform only the inexpensive steady accessibility solve. The closure predicts D_{eff} , and the 1D model then predicts the complete uptake curve. The expensive transient pore-network adsorption calculation belongs to development and validation, not to routine use.

The homogeneous 1D uptake model

Axial diffusion and adsorption

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t},$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$$

For each trial D , solve both equations together for $c(z, t)$ and $q(z, t)$ in 100 cells, with prescribed initial and boundary conditions.

For every trial D , the 1D solver produces a complete mean uptake curve.

Quantity compared with the network

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting.

2026-09-12

- Numerical methods and diffusivity prediction
- The homogeneous 1D uptake model

The homogeneous 1D uptake model

Axial diffusion and adsorption

$$\varepsilon_b \frac{\partial c}{\partial t} = D \frac{\partial^2 c}{\partial z^2} - (1 - \varepsilon_b) \rho_p \frac{\partial q}{\partial t}$$

$$\frac{\partial q}{\partial t} = k_{\text{LDF}} [q^*(c) - q].$$

For each trial D , solve both equations together for $c(z, t)$ and $q(z, t)$ in 100 cells, with prescribed initial and boundary conditions.

Quantity compared with the network

$$\bar{q}_{1D}(t; D) = \frac{1}{H} \int_0^H q(z, t; D) dz.$$

The 1D model uses the same bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state as the pore network. Only D changes during fitting.

For every trial D , the 1D solver produces a complete mean uptake curve.

The reduced model replaces the irregular pore network by one axial coordinate. It solves coupled diffusion and adsorption in one hundred cells. The local outputs are the gas concentration $c(z, t)$ and loading $q(z, t)$. Their spatial average gives $\bar{q}_{1D}(t; D)$, the mean uptake predicted by the 1D model.

The bed height, porosity, inlet condition, Langmuir parameters, LDF coefficient and initial state remain identical to those in the pore-network calculation. Only the diffusivity D changes. For any selected value of D , the solver must integrate the complete transient problem and return a complete uptake curve. This is why we call it a trial diffusivity at this stage. We do not yet know which value best represents the network.

└ Numerical methods and diffusivity prediction

└ D_{access} and D_{eff} answer different questions

D_{access} and D_{eff} answer different questions	
D_{access} : geometric accessibility - fix concentration at inlet and interior plane - solve one steady, linear diffusion problem - no adsorption and no time integration $D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}$ G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.	D_{eff} : transient reduced-model coefficient - full transient network gives $\bar{q}_{\text{PNM}}(t)$ - repeated 1D solves give $\bar{q}_{1D}(t; D)$ Define the curve residual $e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$ $\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$ $D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$ The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$. D_{access} is cheap because it avoids adsorption dynamics and optimization.

D_{access} and D_{eff} answer different questions

D_{access} : geometric accessibility

- fix concentration at inlet and interior plane
- solve one steady, linear diffusion problem
- no adsorption and no time integration

$$D_{\text{access}} = \frac{G_{\text{access}} H}{A_{\text{tube}}}.$$

G_{access} is total steady flux per concentration difference. H is bed height; A_{tube} is tube area.

D_{eff} : transient reduced-model coefficient

- full transient network gives $\bar{q}_{\text{PNM}}(t)$
- repeated 1D solves give $\bar{q}_{1D}(t; D)$

Define the curve residual

$$e_m(D) = \bar{q}_{1D}(t_m; D) - \bar{q}_{\text{PNM}}(t_m).$$

$$\text{NRMSE}(D) = \frac{\sqrt{\sum_m w_m e_m(D)^2}}{\bar{q}_{\text{PNM}}(t_f)},$$

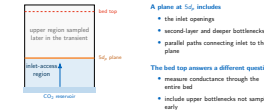
$$D_{\text{eff}} = \arg \min_D \text{NRMSE}(D).$$

The trapezoidal weights w_m account for unequal time spacing and satisfy $\sum_m w_m = 1$.

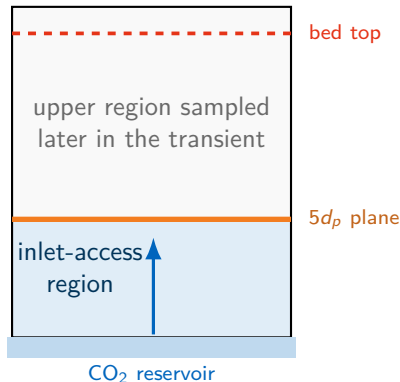
D_{access} is cheap because it avoids adsorption dynamics and optimization.

These two quantities answer different questions. To calculate D_{access} , we prescribe concentrations at the inlet and at an interior plane and solve one steady linear diffusion problem. Steady here means that we directly solve the time-independent balance. We do not run a transient simulation and wait for physical steady state. The result is the total conductance G_{access} between the two boundaries. Multiplying it by bed height and dividing by tube area expresses it in diffusivity units.

D_{eff} is different. It is the coefficient required by the reduced 1D transient model. The network first supplies $\bar{q}_{\text{PNM}}(t)$. We choose a trial D , solve the 1D transient model and obtain $\bar{q}_{1D}(t; D)$. The optimizer then changes D and repeats the 1D solve. At every saved time, $e_m(D)$ is the difference between the 1D and network uptake. NRMSE combines these errors over the full curve. Squaring prevents positive and negative errors from cancelling, the square root restores loading units, and division by the final network loading makes the result dimensionless. The time weights account for unequal intervals. The value that minimizes NRMSE is called D_{eff} . This optimization explains why obtaining D_{eff} is more expensive than calculating D_{access} .



Why use an interior plane?



A plane at $5d_p$ includes

- the inlet openings
- second-layer and deeper bottlenecks
- parallel paths connecting inlet to the plane

The bed top answers a different question

- measure conductance through the entire bed
- include upper bottlenecks not sampled early

The accessibility calculation connects the inlet to an interior plane at a selected depth. It includes every connected path and every bottleneck between these two boundaries. Therefore, a bottleneck in the second layer or any deeper layer below the plane directly reduces G_{access} . We use an interior plane because the descriptor targets access from the inlet into the region sampled during the earlier part of adsorption. A plane at the bed top would instead measure steady conductance through the complete bed. It would include upper restrictions that affect the response mainly later in the transient. This does not mean that upper bottlenecks are physically unimportant. It means that whole-bed conductance answers a different question from inlet accessibility. The appropriate depth must therefore be checked rather than assumed.

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

2026-09-12

└ Numerical methods and diffusivity prediction

└ Why not predict uptake directly from G_{access} ?

Why not predict uptake directly from G_{access} ?

Direct endpoint relation

$$G_{\text{access}} \longrightarrow \bar{q}(5000 \text{ s})$$

- strong correlation: $\rho_s = 0.968$
- ranks beds or predicts one endpoint
- valid at the selected time and conditions

If only uptake at 5000 s is needed, G_{access} is sufficient; D_{eff} is needed for a reusable transient model.

Transport closure

$$G_{\text{access}} \longrightarrow D_{\text{access}} \longrightarrow D_{\text{eff}} \longrightarrow \bar{q}(t)$$

- generates the complete uptake curve
- retains conservative 1D transport
- gives axial $c(z, t)$ and $q(z, t)$ profiles

Accessibility conductance correlates very strongly with uptake at 5000 seconds. If our only task were to rank beds or estimate this one endpoint under the same conditions, we could use G_{access} directly.

Our intended reduced model asks for more. We want the complete uptake curve and the axial concentration and loading profiles. The 1D conservation equations require a transport coefficient, which is D_{eff} . Furthermore, D_{access} should not simply be inserted as D_{eff} because the strict proportional model performs substantially worse than the fitted power-law closure. The sequence is therefore: calculate D_{access} cheaply, predict D_{eff} through the closure, and use D_{eff} in the transient 1D model.

Fit one diffusivity per bed

1. Choose a trial D within prescribed bounds.
2. Solve the coupled 1D equations for $c(z, t)$ and $q(z, t)$.
3. Compare mean uptake with the network using time-weighted NRMSE.
4. Use previous errors to select the next D and repeat (optimization).

Learn the relationship across beds

- Pair each fitted D_{eff} with its geometry-based D_{access} .
- Fit the closure $D_{\text{eff}} = f(D_{\text{access}})$ on the training beds.
- Leave one bed out to assess diffusivity prediction.

The bounded search operates in $\log_{10} D$ and starts deterministically.

Per-bed optimization supplies the targets for closure calibration.

Fit one diffusivity per bed

1. Choose a trial D within prescribed bounds.
2. Solve the coupled 1D equations for $c(z, t)$ and $q(z, t)$.
3. Compare mean uptake with the network using time-weighted NRMSE.
4. Use previous errors to select the next D and repeat (optimization).

Learn the relationship across beds

- Pair each fitted D_{eff} with its geometry-based D_{access} .
- Fit the closure $D_{\text{eff}} = f(D_{\text{access}})$ on the training beds.
- Leave one bed out to assess diffusivity prediction.

The bounded search operates in $\log_{10} D$ and starts deterministically.

Per-bed optimization supplies the targets for closure calibration.

There are two fitting tasks. First, for each bed, optimize one diffusivity to match its network uptake curve. Each trial restarts the coupled transient model from the prescribed initial state. Then learn a relationship between these fitted targets and accessibility across beds. In leave-one-bed-out evaluation, fit that relationship on nineteen beds and predict the twentieth. This is internal validation within the current ensemble.

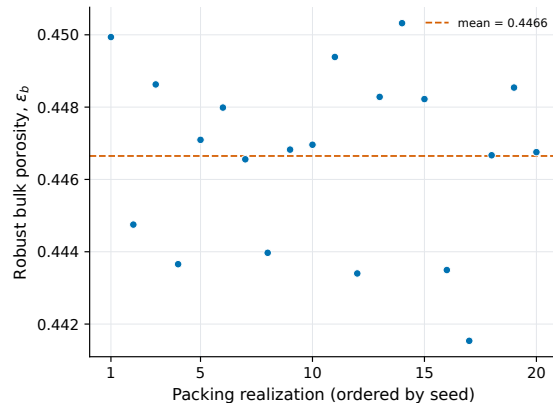
Section 4 of 5

Results and robustness

Uptake variability, predictive closure and depth sensitivity

Uptake variability, predictive closure and depth sensitivity.

The twenty beds have nearly equal bulk porosity



Bulk-porosity CV: 0.55%.

Controlled ensemble

- same bead count and diameter
- same tube geometry
- only the insertion seed changes

CV: coefficient of variation

$$CV_{\epsilon} = s_{\epsilon} / \bar{\epsilon}$$

s_{ϵ} : standard deviation

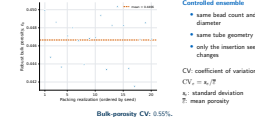
$\bar{\epsilon}$: mean porosity

2026-09-12

Results and robustness

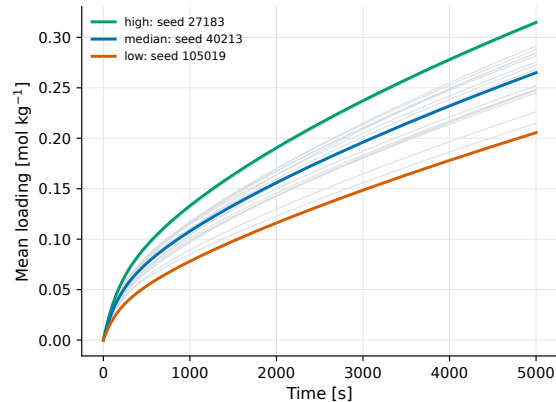
The twenty beds have nearly equal bulk porosity

The twenty beds have nearly equal bulk porosity



The mean bulk porosity is approximately 0.44665. Divide the sample standard deviation by this mean to obtain the coefficient of variation, 0.55 percent. The horizontal axis orders beds by their insertion seed. The small spread establishes the controlled comparison.

Similar porosity still produces different uptake curves



At 5000 s: 0.206–0.315 mol kg⁻¹, with an uptake CV of 11.0%.

Network reference

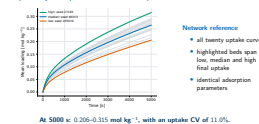
- all twenty uptake curves
- highlighted beds span low, median and high final uptake
- identical adsorption parameters

2026-09-12

Results and robustness

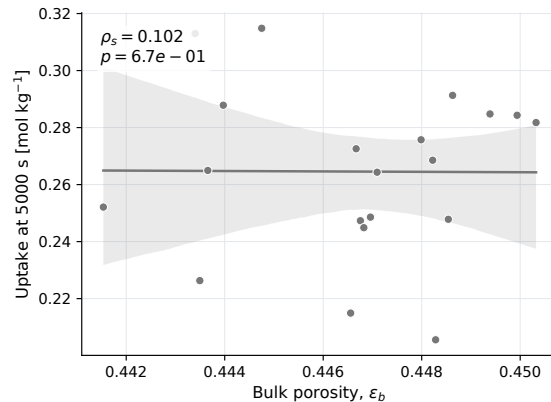
Similar porosity still produces different uptake curves

Similar porosity still produces different uptake curves



The pale curves show the full ensemble. Orange, blue and green identify low, median and high final uptake. These representative beds are selected by final uptake, not by accessibility. The uptake varies much more strongly than porosity.

Bulk porosity alone does not explain final uptake



Porosity baseline

- each point represents one bed
- weak rank association
- wide uncertainty around the fitted relationship

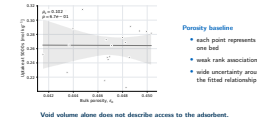
Void volume alone does not describe access to the adsorbent.

2026-09-12

Results and robustness

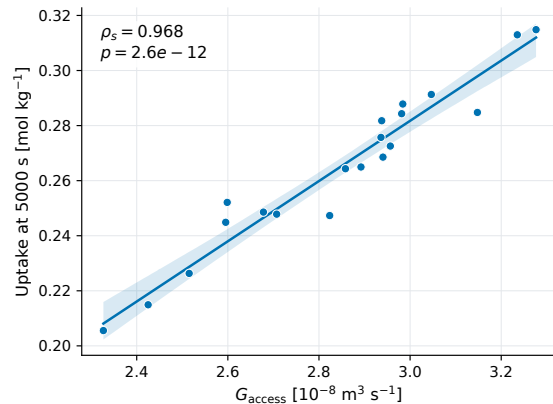
Bulk porosity alone does not explain final uptake

Bulk porosity alone does not explain final uptake



The plot compares bulk porosity with mean loading at 5000 seconds. Spearman correlation is approximately 0.102. The shaded band is pointwise bootstrap uncertainty in the fitted relationship, not a prediction interval for a new bed.

Accessibility explains differences in final uptake



Inlet-connected pathways explain variability that porosity misses.

Steady network descriptor

- G_{access} measures inlet-to-interior conductance
- greater conductance accompanies greater uptake

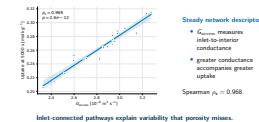
Spearman $\rho_s = 0.968$.

2026-09-12

Results and robustness

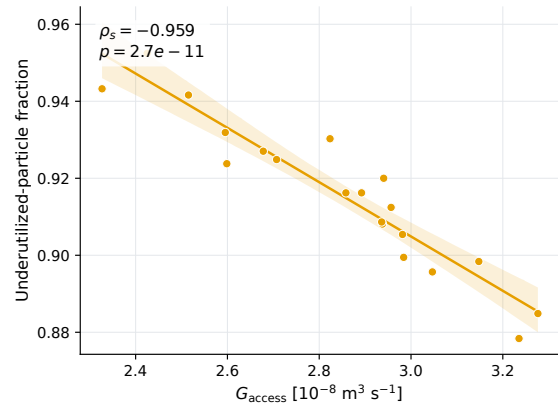
Accessibility explains differences in final uptake

Accessibility explains differences in final uptake



We calculate accessibility using a steady diffusion problem without adsorption. The Spearman coefficient is 0.968. More accessible beds generally have greater uptake. This is a within-ensemble association. The shaded band represents uncertainty in the fitted relationship.

Greater accessibility reduces particle underutilization



Higher G_{access} accompanies fewer underutilized particles.

Underutilized fraction

$$f_{\text{under}} = \frac{1}{N_p} \sum_{k=1}^{N_p} I_k.$$

$I_k = 1$ when

$$q_k(t_f) < 0.5q^*(p_{\text{inlet}}).$$

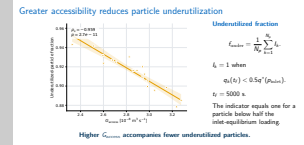
$t_f = 5000 \text{ s}.$

The indicator equals one for a particle below half the inlet-equilibrium loading.

2026-09-12

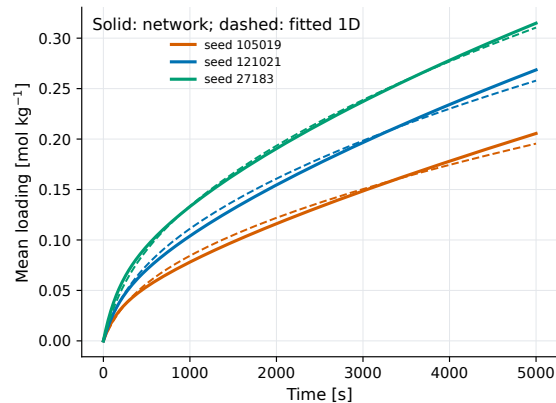
Results and robustness

Greater accessibility reduces particle underutilization



For each bead we compare its final loading with half the equilibrium loading at the inlet pressure. The indicator is one below this threshold and zero otherwise. Averaging over particles gives the plotted fraction. This is a threshold-based utilization measure, not the fraction of completely inaccessible particles.

One fitted D_{eff} reproduces each uptake curve



Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

Per-bed optimization

1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

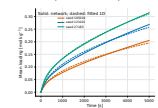
Solid: network.
Dashed: fitted 1D model.

2026-09-12

Results and robustness

One fitted D_{eff} reproduces each uptake curve

One fitted D_{eff} reproduces each uptake curve



Per-bed optimization

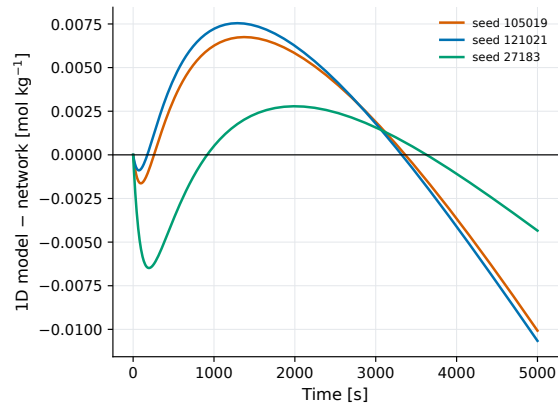
1. solve the 1D model for a trial D
2. evaluate time-weighted NRMSE against the network curve
3. use previous errors to select the next D and repeat (optimization)

Solid: network.
Dashed: fitted 1D model.

Across twenty beds, the mean fitted-curve NRMSE is 1.58%.

For each bed, the transient pore-network calculation provides a reference uptake curve. We repeatedly solve the 1D model with trial diffusivities and minimize the time-weighted normalized root-mean-square error. The bounded optimizer searches in $\log_{10} D$. Its initial trial follows from the prescribed bounds, without random sampling. Previous objective values guide subsequent trials until the search meets its tolerance. Each trial solves both coupled equations from the same initial state. The optimized diffusivity is D_{eff} . This figure shows per-bed fits, not predictions for unseen beds. The mean error is 1.58 percent.

Residuals show the remaining reduction error



Time-weighted NRMSE across all beds: 0.84–2.46%.

Curve residual

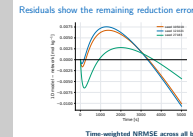
$$e(t) = \bar{q}_{1D}(t) - \bar{q}_{PNM}(t).$$

- positive: excess predicted uptake
- negative: insufficient predicted uptake

2026-09-12

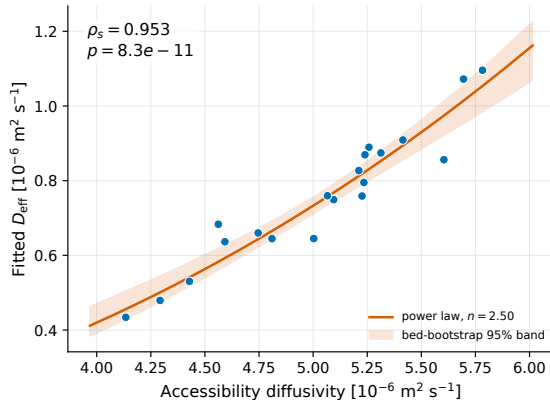
Results and robustness

Residuals show the remaining reduction error



The same three representative beds appear here. The residual is the 1D mean loading minus the network mean loading. Its sign changes over time, showing that one coefficient cannot reproduce every detail. NRMSE summarizes the full residual curve using trapezoidal time weights and normalizes by final network loading.

A positive power law relates accessibility to diffusivity



The fitted exponent is $n \approx 2.50$ for this ensemble.

Selected closure

$$\frac{D_{eff}}{D_m} = C \left(\frac{D_{access}}{D_m} \right)^n$$

$$D_{access} = \frac{G_{access} H}{A_{tube}}$$

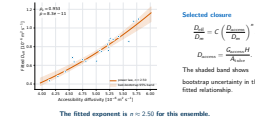
The shaded band shows
bootstrap uncertainty in the
fitted relationship.

2026-09-12

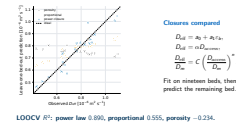
Results and robustness

A positive power law relates accessibility to diffusivity

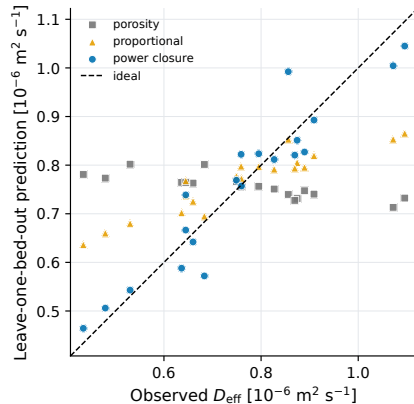
A positive power law relates accessibility to diffusivity



Each bed supplies a geometrically calculated accessibility diffusivity and a diffusivity fitted to its transient response. The power law relates them. This plot shows the fit to the complete ensemble. The band is pointwise bootstrap uncertainty in the relationship, not an individual-bed prediction interval. The exponent is specific to the studied conditions.



The closure predicts diffusivity for a withheld bed



Closures compared

$$D_{\text{eff}} = a_0 + a_1 \varepsilon_b,$$

$$D_{\text{eff}} = \alpha D_{\text{access}},$$

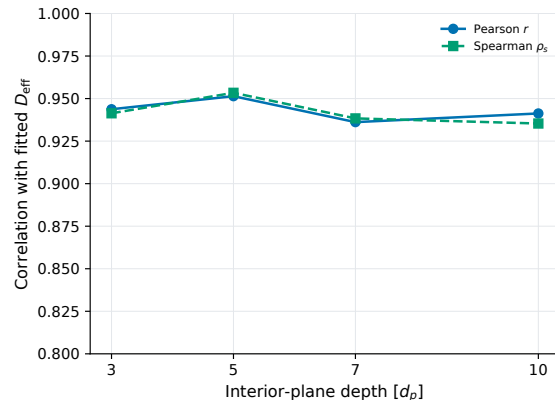
$$\frac{D_{\text{eff}}}{D_m} = C \left(\frac{D_{\text{access}}}{D_m} \right)^n.$$

Fit on nineteen beds, then predict the remaining bed.

LOOCV R^2 : power law 0.890, proportional 0.555, porosity -0.234.

We leave out one entire bed, fit each closure to the other nineteen, and predict the omitted diffusivity. Repeating this gives twenty held-out predictions. Blue circles show the power law, orange triangles show proportionality, and grey squares show porosity. Model form and diagnostic depth were compared within this ensemble, so these scores remain internal validation.

Accessibility remains correlated across diagnostic depths



Both correlations remain high across the tested depths.

Depth sensitivity

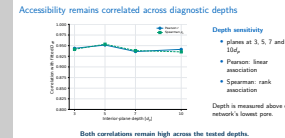
- planes at 3, 5, 7 and $10d_p$
- Pearson: linear association
- Spearman: rank association

Depth is measured above each network's lowest pore.

2026-09-12

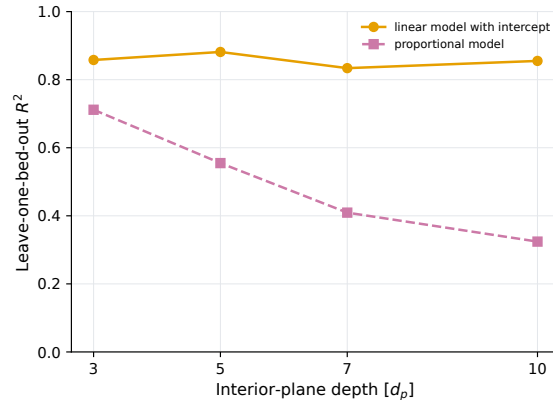
Results and robustness

Accessibility remains correlated across diagnostic depths



The original descriptor places the diagnostic plane above the lowest pore in each network. We tested three, five, seven and ten particle diameters using that convention. Both linear and rank correlations remain high. These depths are not measured exactly from the physical bed bottom.

Predictive performance is stable across diagnostic depths



Models in this comparison

- linear model with intercept

$$D_{\text{eff}} = aD_{\text{access}} + b$$

- proportional model

$$D_{\text{eff}} = \alpha D_{\text{access}}$$

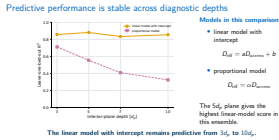
The $5d_p$ plane gives the highest linear-model score in this ensemble.

The linear model with intercept remains predictive from $3d_p$ to $10d_p$.

2026-09-12

Results and robustness

Predictive performance is stable across diagnostic depths



This plot compares affine and proportional closures at each depth. It does not plot the power-law score. Five particle diameters gives the best linear-model score among these tested depths. This sensitivity supports the association while independent beds are still needed to evaluate model and depth selection.

Section 5 of 5

Conclusions and next steps

What the evidence supports and what remains to validate

What the evidence supports and what remains to validate.

What the present evidence establishes

Supported

- porosity alone misses important packing variability
- inlet-connected topology controls transient utilization
- one D_{eff} reproduces each detailed curve accurately
- D_{access} provides an interpretable predictive closure

Not yet claimed

- industrial breakthrough performance
- effects of flow, heat or water competition
- transfer to other bead sizes or D_{tube}/d_p
- a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

2026-09-12

└─ Conclusions and next steps

└─ What the present evidence establishes

What the present evidence establishes

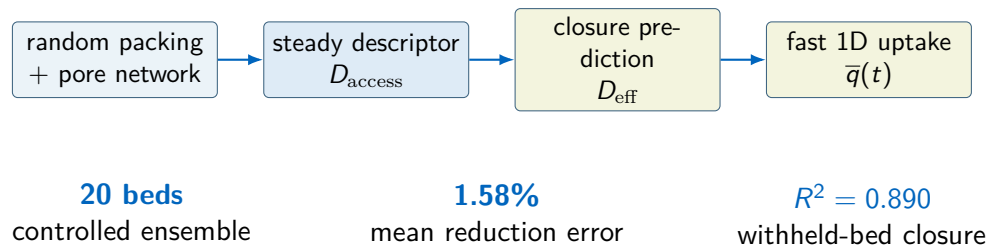
Supported	Not yet claimed
• porosity alone misses important packing variability • inlet-connected topology controls transient utilization • one D_{eff} reproduces each detailed curve accurately • D_{access} provides an interpretable predictive closure	• industrial breakthrough performance • effects of flow, heat or water competition • transfer to other bead sizes or D_{tube}/d_p • a universal power-law exponent

The contribution is a structure-to-model closure for a controlled diffusion-driven adsorption problem.

The present evidence supports four conclusions. Bulk porosity misses important differences among random packings. Inlet-connected topology explains a large part of the resulting variation. One optimized D_{eff} allows the 1D model to reproduce each detailed uptake curve accurately. Finally, D_{access} provides a cheap and interpretable input for predicting that coefficient. The scope must remain clear. We have not modelled an industrial breakthrough column with imposed flow. We have not included heat effects, water or competitive adsorption. We also have not demonstrated that the fitted power law transfers to other bead sizes or tube-to-particle ratios. The result is a validated structure-to-model closure for this controlled, diffusion-driven adsorption problem, rather than a universal adsorption correlation.

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



2026-09-12

Conclusions and next steps

Pore accessibility retains what bulk porosity loses

Pore accessibility retains what bulk porosity loses

Equal void volume does not imply equal transport. How that void space connects to the inlet matters.



The main message is that equal void volume does not imply equal transport. How the void space connects to the inlet matters. Starting from a random packing, we extract the pore network and calculate the steady accessibility descriptor. The fitted closure converts this descriptor into D_{eff} , and the conservative 1D model uses that coefficient to predict the transient uptake.

Across twenty controlled beds, porosity varied by only 0.55 percent while uptake varied much more strongly. The optimized 1D model reproduced the pore-network curves with a mean NRMSE of 1.58 percent. The accessibility power law achieved a leave-one-bed-out R^2 of 0.890. The practical outcome is therefore a fast transient model that retains a clear and measurable link to the underlying packing structure.